

Experiment + Sensing Protocol

Scientific rules behind the consumer experience.

VERSION	STATUS	TARGET
v0.3	Research / engineering protocol	OpenCode / implementation team

Purpose: This document separates what is fun in the UI from what must stay controlled in the experiment. It is designed to prevent the product from producing a visually convincing but scientifically meaningless "winner."

1. Primary experimental question

Within the same user and session, are repeated exposures to particular digital color/theme conditions associated with reliable changes in observable physiological and facial-response signals relative to that user's session baseline?

Primary endpoints for the MVP

- Heart-rate change from session baseline.
- Respiration-rate change when the respiration estimator is valid.
- Camera-derived pulse variability / PRV trend when enough clean beats exist.
- Signal-quality coverage and motion/illumination stability.
- Calibrated facial-expression tendency as a secondary signal, not a direct readout of internal emotion.

Interpretation rule: Lower HR is not automatically "better" and higher HR is not automatically "stress." Separate calm-associated response, positive activation and stress-like activation.

2. Why repeated randomization is required

A fixed sequence such as Blue -> Green -> Red makes condition inseparable from time/order. Physiology can drift because of adaptation, fatigue, posture, attention, model stabilization and random variation. Therefore every candidate must be repeated and the order must change.

Recommended quick-mode design

Phase	Conditions	Timing	Notes
Baseline	Neutral	25-30 s	Natural breathing; no talking; camera visible
Screening A	5 colors	12 s each	Random order; first ~3 s excluded
Washout	Neutral	~4 s	Between color trials
Screening B	Same 5 colors	12 s each	New random order
Finalists	Top 2 colors	~30 s each	Randomized confirmation + neutral washout
Chat-theme confirmation	2-4 themes	~15-20 s each	Real chat environment; neutral content
Optional font	5 font categories	~10-12 s each x repeats	Color held constant
Optional emoji	5 emoji states	~10 s each x repeats	Color/font held constant

This is a consumer protocol. It is intended to detect strong repeatable within-session tendencies, not to replace laboratory-quality long-duration HRV studies.

3. Trial-state requirements

- Persist a randomization seed and exact condition order.
- Each candidate receives the planned number of repeats.
- No two identical conditions occur adjacent unless intentionally part of a finalist confirmation.
- Mark the first about 3 seconds after every screen change as transition.
- Do not analyze rejected/low-SQI samples as if they were valid.
- If a condition has insufficient valid duration, reschedule/repeat it or mark it inconclusive.
- Do not force a winner if repeat directions conflict.

Suggested data record

```
{ trial_id, condition_type, condition_id, repeat_index, randomization_seed, t_start, t_transition_end, t_end,
  hr_mean, hr_delta, rr_mean, rr_delta, rmssd_prv, expression_vector, motion_score, sqi_coverage, frame_luminance,
  valid_seconds, rejection_reason }
```

4. Baseline rules

Baseline exists to create a within-session reference, not to relax the participant as much as possible.

Correct user instruction: Look comfortably at the screen. Breathe normally. Blink normally. Try not to talk or move too much.

- Do not ask the participant to look away.
- Do not use slow-breathing coaching or paced breathing.
- Do not show emotionally charged content.
- Use a neutral, approximately mid-lightness visual background.
- Require adequate signal quality before the baseline clock begins.
- Keep the mirrored camera and signal-quality indicator visible.

5. Screen color is also a light source

This is the central scientific risk of EmOcean AI. rPPG works from tiny color changes in reflected facial light. When the display changes hue, the display can change the spectrum and intensity of light hitting the face. An apparent color effect may therefore be partly optical rather than physiological.

Minimum mitigation

- Choose experimental colors in a perceptual space such as OKLCH and approximately match lightness.
- Keep device brightness unchanged throughout a session where practical.
- Record the exact presented color and timestamps.
- Record frame/ROI luminance or other illumination-quality measures.
- Use neutral washouts and exclude transition windows.
- Flag trials with large illumination discontinuities.
- Keep an independent rPPG estimator available for comparison if possible.
- Validate internally with a synchronized contact PPG while screen colors change.

Go/no-go: If display hue changes produce apparent camera HR changes while contact PPG remains stable, do not make physiological color claims until the correction pipeline is improved.

6. Signal hierarchy

Signal	Role	Caveat
Heart rate	Primary consumer signal	Needs SQI; movement/light can create error
BVP waveform	Underlying rPPG signal	Do not expose raw waveform as clinical PPG
Beat intervals	Supports PRV	Needs clean beats; reject noisy windows
RMSSD/SDNN-like outputs	Secondary PRV trend	Call camera-derived pulse variability / PRV, not ECG-equivalent HRV
Respiration estimate	Secondary primary endpoint	Needs confidence; face-rPPG alone may fail during motion/talking
Facial expression	Secondary affect channel	Expression is not internal emotion
Gaze / head pose	Quality + behavioral context	Useful for "looking away" and motion rejection
Mouth/speaking	Quality exclusion	Talking strongly affects motion/respiration estimates
SQI	Mandatory quality gate	Do not return confident physiology without it

7. Respiration implementation

For the MVP, estimate respiration frequency from low-frequency modulation of the rPPG/BVP signal with a confidence value and a rejection reason. Next, add a second channel based on subtle upper-torso/shoulder motion using Pose Landmarker or optical-flow-style motion. Fuse only when the channels are individually valid and reasonably agree.

Do not overclaim: A camera-derived respiration-rate estimate is feasible. "Deepest breathing" is a separate measurement claim; absolute breath depth should not be presented until validated against a respiratory belt/reference.

8. Color -> chat -> font -> emoji separation

Do not change several visual dimensions at once if the goal is to learn which dimension matters.

Module	Change	Hold constant
Color Currents	Background hue/lightness-controlled stimulus	Font, chat content, bubble layout
Chat Tides	Full background + bubble palette derived from finalists	Font and neutral content structure
Typography	Font category	Color/theme
Emoji Mirror	Emoji/expression cue	Color, font, message content
Final combination	Best candidate dimensions	Only after individual ranking

9. Neutral chat-content rules

- Avoid positive/negative news, arguments, romantic content, money, health scares or other emotionally loaded topics.
- Match approximate word count and reading difficulty.
- Match number/length of chat bubbles and message pacing.
- Randomize neutral text-set assignment independently of visual condition.
- Do not ask the user to rate each condition while it is being measured; ratings can change attention and expectation.

Examples

"Did you leave the notebook on the second shelf?" / "Yes, beside the blue folder." "What time does the cafe open?" / "Eight, according to the sign."

10. Scoring framework

Start with transparent, inspectable within-session scoring rather than a black-box "emotion" label.

Example hypothesis score

```
calm_association = 0.45*(-z(HR_delta)) + 0.30*(-z(RR_delta)) + 0.15*z(PRV_delta) + 0.10*z(face_valence_delta)
```

The weights above are an MVP hypothesis, not a validated clinical biomarker. Disable/down-weight PRV when too few clean beats exist. Add or remove facial-affect weighting based on calibration quality.

Activation

Higher HR and/or respiration with neutral-positive facial response can be labelled activation/energizing rather than stress.

Stress-like

Only use stress-like language when multiple independent features point in the same direction, such as HR up + respiration up + PRV down + negative facial-response tendency, with adequate quality.

11. Confidence and uncertainty

- Signal-quality coverage per condition.
- Valid analysis seconds after transition exclusion.
- Number of clean beats for PRV.
- Motion and speaking contamination.
- Illumination stability.
- Agreement between repeat A and repeat B.
- Margin between first and second candidate.
- Agreement across independent signals.
- If available, agreement between primary and secondary rPPG estimators.

Required behavior: When evidence is weak, return "We did not get a clear winner this time" and offer a retry. A scientifically modest uncertain result is better than a falsely precise winner.

12. iPhone / browser measurement QA

Test	Pass criterion
Physical iPhone Safari camera	Permission from tap; mirrored preview starts and stays alive
Stage transitions	No video element/stream remount or black frame
Background / foreground	Recover gracefully after returning to Safari
Low light	Warn and pause/reject rather than output fake confidence
Head turn / talking	Quality falls or trial is rejected appropriately
Model load	Progress shown; timeout/retry exists
Viewport	No controls hidden by safe areas/browser chrome
Thermal / long session	No runaway CPU/memory; frame processing remains responsive

13. Validation before public accuracy claims

Stage	Objective
A. Device smoke test	Confirm permissions, camera lifecycle, model load, PWA/mobile behavior.
B. Sensor validity	Synchronized webcam + contact PPG across devices, skin tones, lights and mild head movement. Report error and rejection rate.
C. Color-confound test	Change display colors while contact PPG verifies true pulse; quantify optical bias.
D. Repeatability	Repeat the same user on separate sessions/days; assess ranking stability.
E. Product-effect study	Only after A-D, test personalized winning theme vs control for replicable response.

Marketing gate: Do not advertise a specific BPM accuracy or claim that a color "reduces stress" until the relevant validation has actually been reproduced on the shipping devices and protocol.

14. Reference links

Vital Camera SDK: <https://github.com/KegangWangCCNU/vitalcamera-sdk>

FacePhys demo: <https://github.com/KegangWangCCNU/FacePhys-Demo>

Face Landmarker - Web: https://ai.google.dev/edge/mediapipe/solutions/vision/face_landmarker/web_js

MediaPipe web setup: https://ai.google.dev/edge/mediapipe/solutions/setup_web

MDN getUserMedia: <https://developer.mozilla.org/en-US/docs/Web/API/MediaDevices/getUserMedia>

Dynamic-lighting rPPG paper: <https://arxiv.org/abs/2507.04306>

Bottom line: The fastest useful consumer experiment is short, randomized, repeated and quality-gated. The most important scientific work is proving that display color changes the person more than it changes the optical rPPG measurement.